教育部課程改進計畫
人工智慧技術與應用 - 最先進之電腦對局遊戲系統設計

Chap. 8 Logical agent

課程：人工智慧導論

授課教師：周信宏

“Thinking Rationally”

- Computational models of human “thought” processes
- Computational models of human behavior
- Computational systems that “think” rationally
- Computational systems that behave rationally

Logical Agents

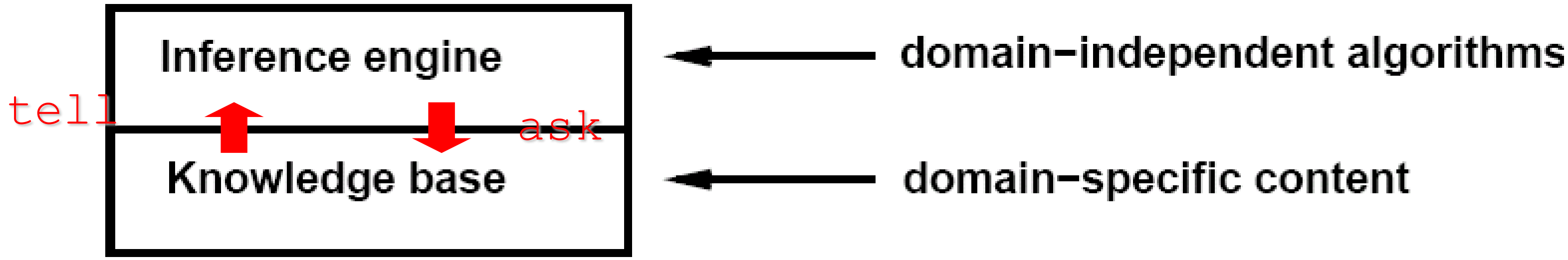
- Reflex agents find their way from Arad to Bucharest by dumb luck (not logical)
- Chess program calculates legal moves of its king, but doesn't know that no piece can be on 2 different squares at the same time (not logical)
- Logic (Knowledge-Based) agents combine general knowledge with current percepts to infer hidden aspects of current state prior to selecting actions
  - Crucial in partially observable environments

Outline

- Knowledge-based agents
- Wumpus world
- Logic in general
- Propositional and first-order logic
  - Inference, validity, equivalence and satifiability
  - Reasoning patterns
    - Resolution
    - Forward/backward chaining

Knowledge Base

Knowledge Base : set of sentences represented in a knowledge representation language and represents assertions about the world.



Inference rule: when one ASKs questions of the KB, the answer should *follow* from what has been TELLed to the KB previously.

Generic KB-Based Agent

```
function KB-AGENT(percept) returns an action
    static: KB, a knowledge base
            t, a counter, initially 0, indicating time

    TELL(KB, MAKE-PERCEPT-SENTENCE(percept, t))
    action ← ASK(KB, MAKE-ACTION-QUERY(t))
    TELL(KB, MAKE-ACTION-SENTENCE(action, t))
    t ← t + 1
    return action
```

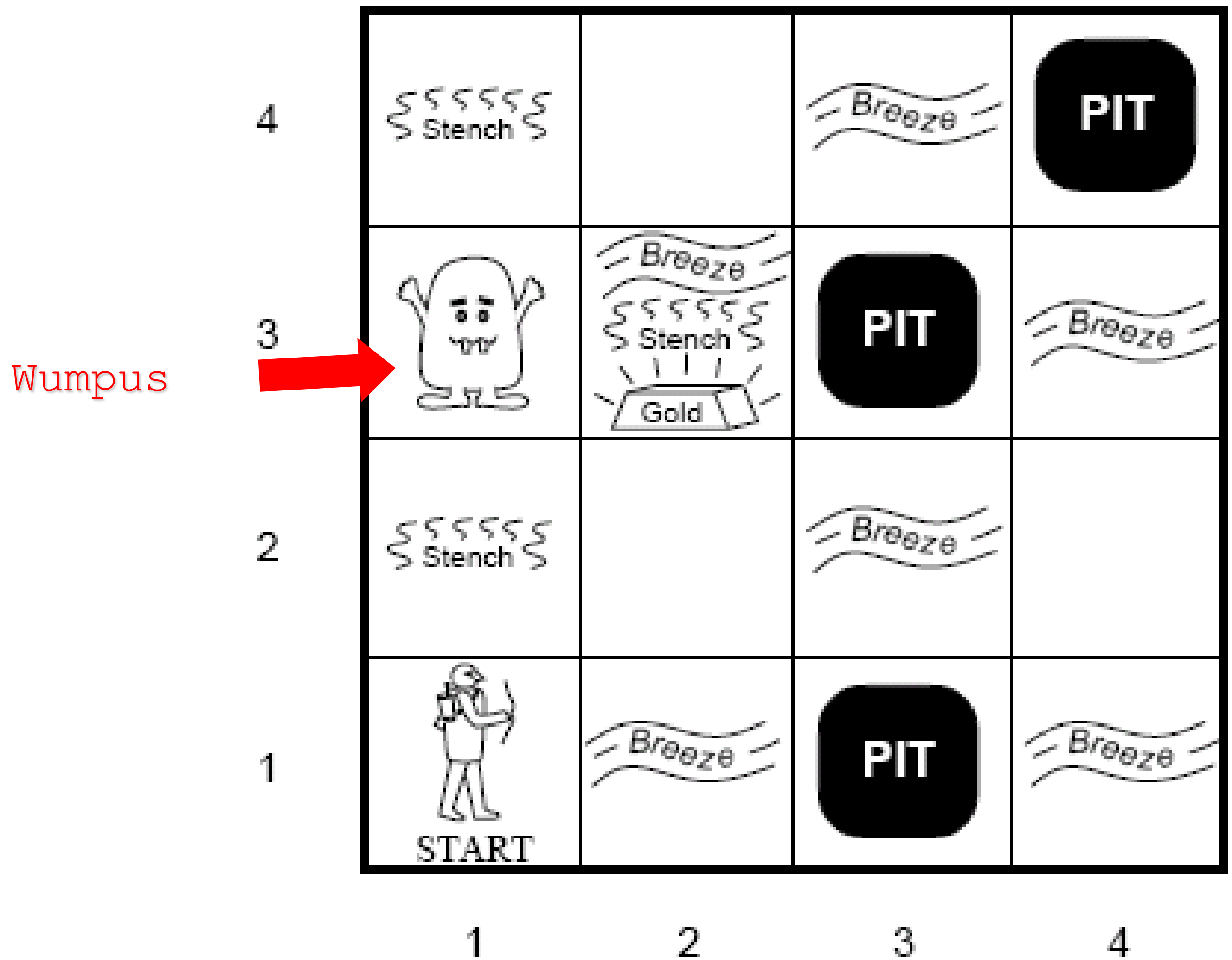
Abilities KB agent

- Agent must be able to:
  - Represent states and actions,
  - Incorporate new percepts
  - Update internal representation of the world
  - Deduce hidden properties of the world
  - Deduce appropriate actions

Desription level

- The KB agent is similar to agents with internal state
- Agents can be described at different levels
  - Knowledge level
    - What they know, regardless of the actual implementation. (Declarative description)
  - Implementation level
    - Data structures in KB and algorithms that manipulate them e.g propositional logic and resolution.

A Typical Wumpus World



Wumpus World PEAS Description

Performance measure

gold +1000, death -1000

-1 per step, -10 for using the arrow

Environment

Squares adjacent to wumpus are smelly

Squares adjacent to pit are breezy

Glitter iff gold is in the same square

Shooting kills wumpus if you are facing it

Shooting uses up the only arrow

Grabbing picks up gold if in same square

Releasing drops the gold in same square

Sensors Breeze, Glitter, Smell

Actuators Left turn, Right turn,
Forward, Grab, Release, Shoot

Wumpus World Characterization

- Observable?
- Deterministic?
- Episodic(情境式)?
- Static?
- Discrete?
- Single-agent?

Wumpus World Characterization

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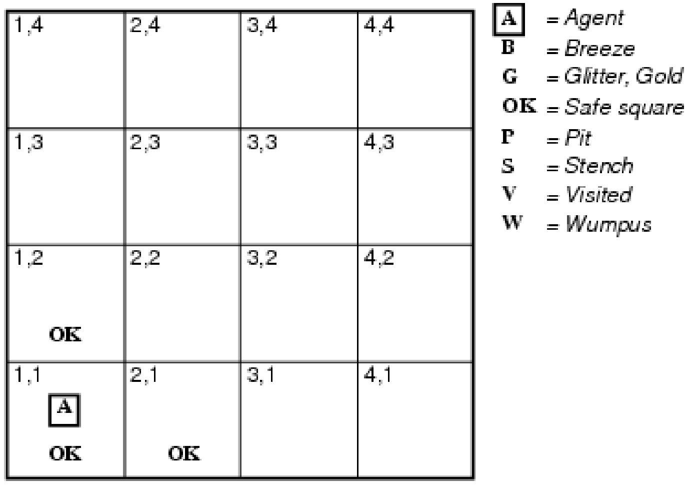
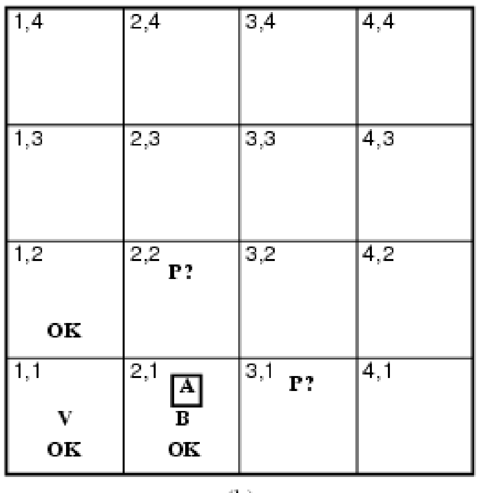
- Discrete? Yes
- Single-agent? Yes, Wumpus is essentially a natural feature.

Exploring the Wumpus World



(a)

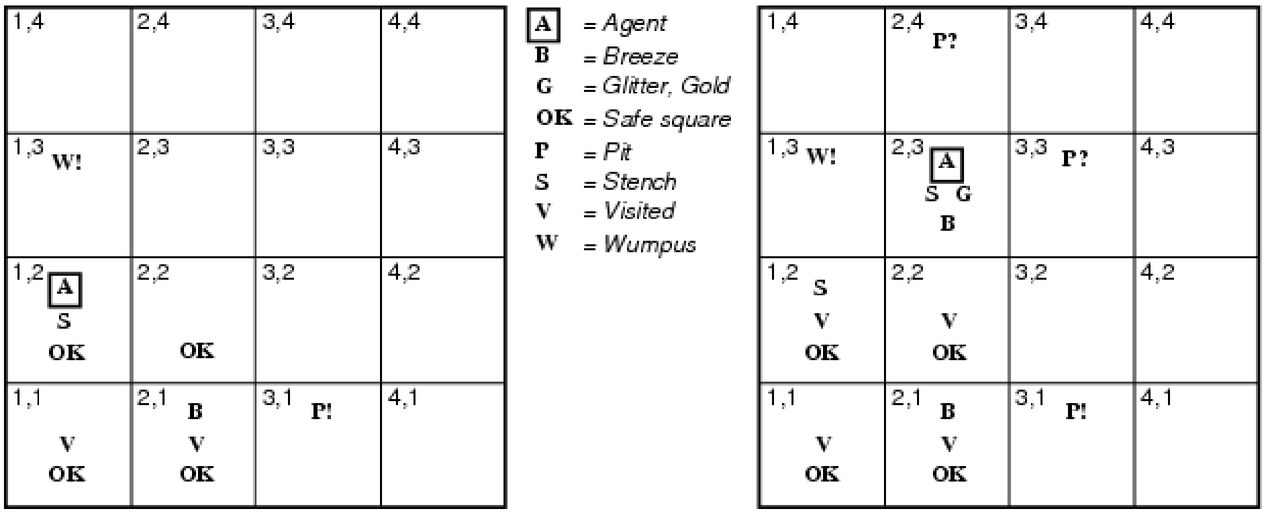


(b)

[1,1] The KB initially contains the rules of the environment. The first percept is [*none, none,none,none,none*], move to safe cell e.g. 2,1

[2,1] breeze which indicates that there is a pit in [2,2] or [3,1], return to [1,1] to try next safe cell

Exploring the Wumpus World



(a) (b)

[1,2] Stench in cell which means that wumpus is in [1,3] or [2,2]

YET … not in [1,1]

YET … not in [2,2] or stench would have been detected in [2,1]

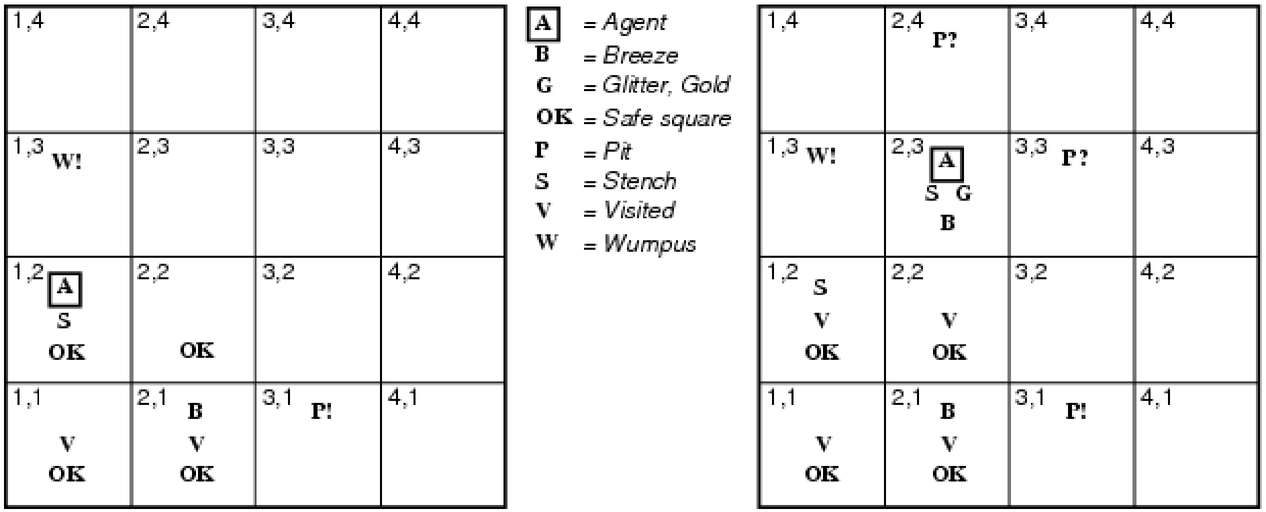
THUS … wumpus is in [1,3]

THUS [2,2] is safe because of lack of breeze in [1,2]

THUS pit in [1,3]

move to next safe cell [2,2]

Exploring the Wumpus World



(a) (b)

[2,2] move to [2,3]
[2,3] detect glitter , smell, breeze
  THUS pick up gold
  THUS pit in [3,3] or [2,4]

What is a logic?

- A formal language
  - Syntax – what expressions are legal (well-formed)
  - Semantics – what legal expressions mean
- In logic, the truth of each sentence with respect to each possible world.
- E.g the language of arithmetic
  - X+2 >= y is a sentence, x^2+y is not a sentence
  - X+2 >= y is true in a world where x=7 and y =1
  - X+2 >= y is false in a world where x=0 and y =6

Entailment (蘊含關係)

- One thing follows from another

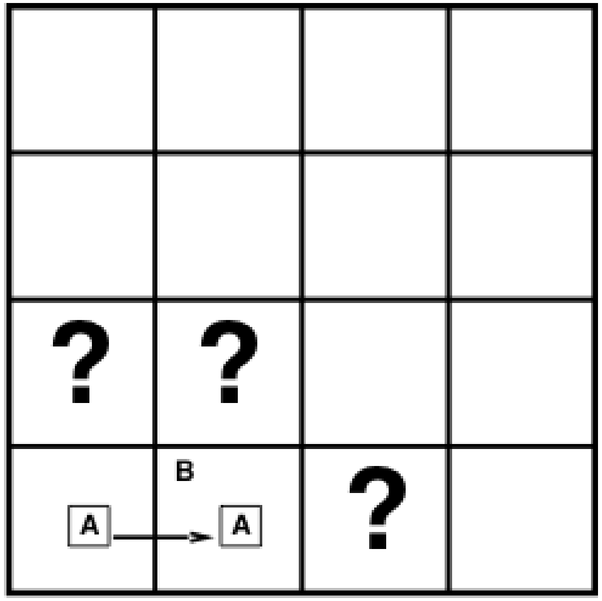
  $KB \models \alpha$

- KB entails sentence α *if and only if* α is true in worlds where KB is true.
- E.g. x+y=4 entails 4=x+y
- Entailment is a relationship between sentences that is based on semantics.

Models

- Logicians typically think in terms of models, which are formally structured worlds with respect to which truth can be evaluated.
- m is a model of a sentence α if α is true in m
- $M(\alpha)$ is the set of all models of α

Wumpus world model

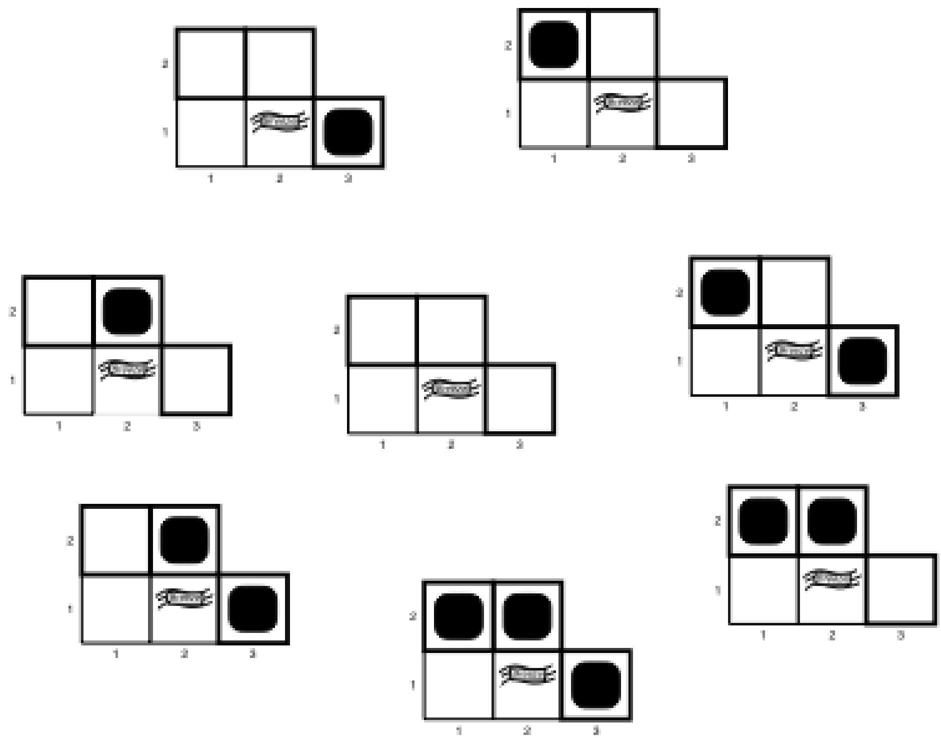


Situation after detecting nothing in [1,1], moving right, breeze in [2,1]

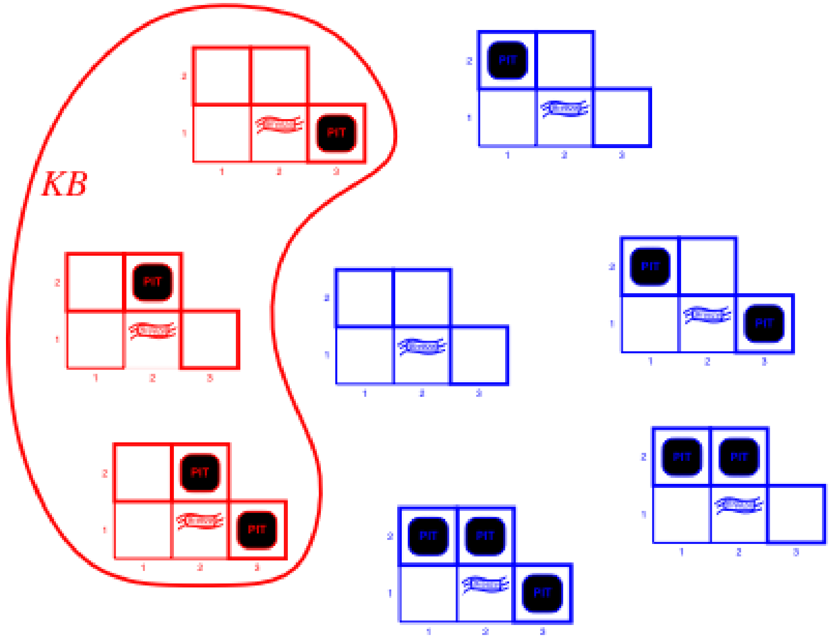
Consider possible models for ?s assuming only pits

3 Boolean choices $\Rightarrow$ 8 possible models

Wumpus world model

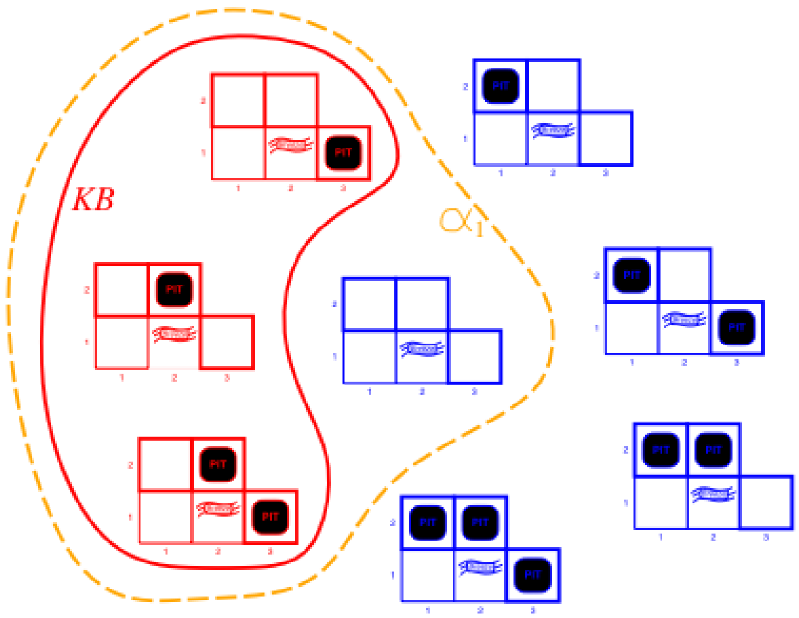


Wumpus world model



KB = wumpus-world rules + observations

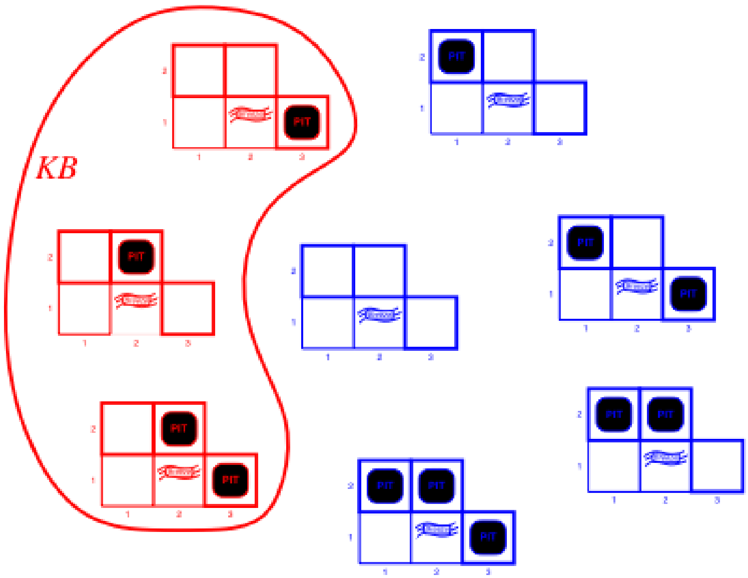
Wumpus world model



KB = wumpus-world rules + observations

α_1 = "[1,2] is safe", $KB \models \alpha_1$, proved by model checking

Wumpus world model



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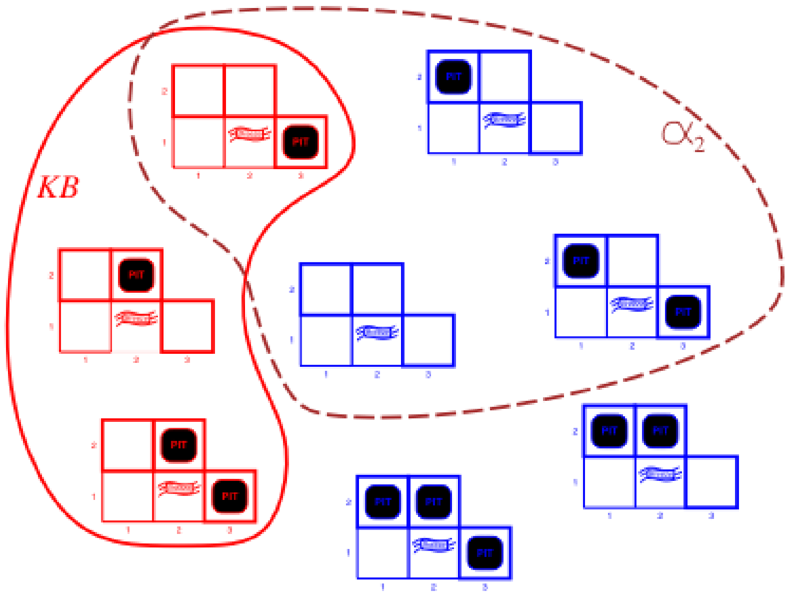


KB = wumpus-world rules + observations

α_2 = "[2,2] is safe", $KB \not\models \alpha_2$

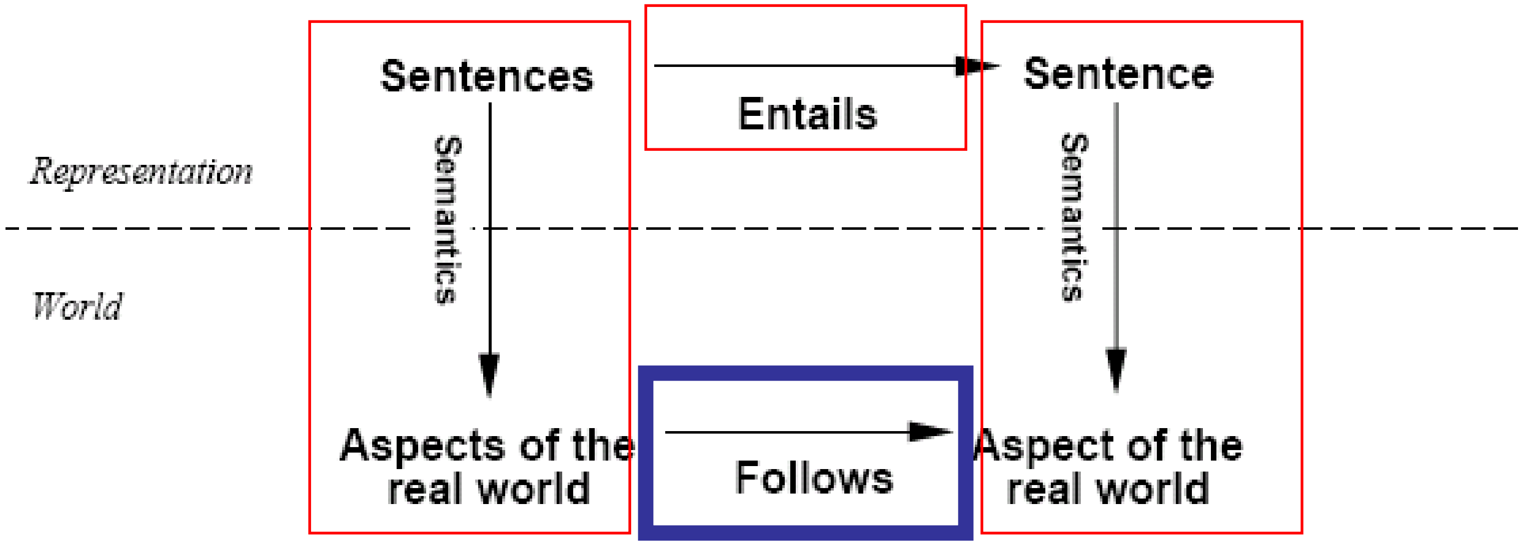
Logical inference (推理)

- The notion of entailment can be used for logic inference.
  - Model checking (see wumpus example): enumerate all possible models and check whether α is true.
- If an algorithm only derives entailed sentences it is called *sound* or *truth-preserving*.
  - Otherwise it just makes things up.

  i is sound if whenever KB $\vdash_i \alpha$ *it is also true that KB* $\models \alpha$
- Completeness : the algorithm can derive any sentence that is entailed.

  i is complete if whenever KB $\models \alpha$ *it is also true that KB* $\vdash_i \alpha$

Schematic perspective



If KB is true in the real world, then any sentence α derived From KB by a sound inference procedure is also true in the real world.

THE END.